



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

SCHOOL OF OPEN, DISTANCE AND eLEARNING

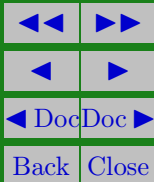
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ICS 2302 SOFTWARE ENGINEERING 1

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 3

Phases in The Development of Software

Learning Outcomes

A study of this lesson should enable students to:

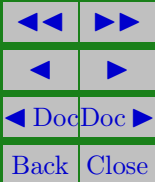
- Define software development life cycle.



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There are a distinguishable number of phases in the development of software. These phases are as depicted in the figure 3.1 below.

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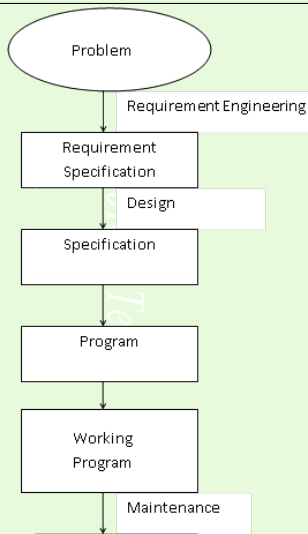


Figure 3.1:



In figure 3.1 the phases have been depicted sequentially. For a given project these activities are not necessarily separated as strictly as indicated here. They may and usually will overlap. It is for instance, quite possible to start implementation of one part of the system while some of the other parts have not been full designed yet.

There is no strict linear progression from requirements engineering to design, from design to implementation etc. back tracking to earlier phases occurs, because of errors discovered or changing requirements.

In the following sections, a short description is given of each of the basic components from figure 3.1.



3.1. Requirements Engineering

The goal of the requirements engineering phase is to get a complete description of the problem to be solved and the requirements posed by and on the environment in which the system is going to function. Requirements posed by the environment may include hardware and supporting software or the number of prospective users of the system to be developed.

Alternatively, analysis of the requirements may lead to certain constraints imposed on hardware yet to be acquired or to the organization in which the system is to function. A description of the problem to be solved includes such things as:

- The functions of the software to be developed.
- Possible future extensions to the system



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- The amount, and kind of documentation required
- Response time and other performance requirements of the system.

Part of the requirements engineering is a feasibility study. The purpose of the feasibility study is to assess whether there is a solution to the problem which is both economically and technically feasible.

The more careful we are during the requirements engineering phase, the larger the chance that the ultimate system will meet expectations. To this end, the various people (among others, the customer, prospective users, designers and programmers) involved have to collaborate intensively.

The document in which the result of this activity is laid down is called the requirements specification.



3.2. Design

During the design phase, a model of the system is developed which, when encoded in some programming language, solves the problem for the user. To this end, the problem is decomposed into manageable pieces called modules or components; the functions of these modules and the interfaces between them are specified in very precise way. The design phase is crucial. During the design phase we try to separate the what from the how. We concentrate on the problem and should not let ourselves be distracted by implementation concerns. The result of this design phase is the (technical) specification, serves as a starting point for the implementation phase.



3.3. Implementation

During the implementation, we concentrate on the individual modules. Our starting point is the modules specification. It is important to note that the first goal of a programmer should be the development of a well documented, reliable easy to read, flexible, correct program. The result of the implementation phase is an executable program.

3.4. Testing

It is wrong to say that testing is a phase following implementation. This suggests that you need not bother testing until implementation is finished.

Attention has to be paid to testing even during the requirements engineering phase. During the subsequent phases, testing



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is continued and refined. The earlier that error's are detected the cheaper it is to correct them. Testing at phase boundaries comes in two flavors. We have to test that the transition between subsequent phases is correct (this is known as verification). We also have to check that we are still on the right track as regards fulfilling user requirements (validation). The result of adding verification and validation activities to the linear model of figure 3.1 yields the so called waterfall model of software development.

3.5. Maintenance

After delivery of the software, there are often errors that have still gone undetected. These errors must be repaired. In addition, the actual use of the system can lead to requests for changes and enhancements.



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Maintenance thus concerns all activities needed to keep the system operational after it has been delivered to the user. An activity spanning all phases is project management. Like other projects, software development projects must be managed properly in order to ensure that the product is delivered on time and within budget.

An important activity not identified separately is documentation. Key components of system documentation include the project plan, quality plan, requirements specification, architecture description, design documentation and test plan.

Figure 3.2 depicts the relative effort spent on the various activities up to delivery of the system.



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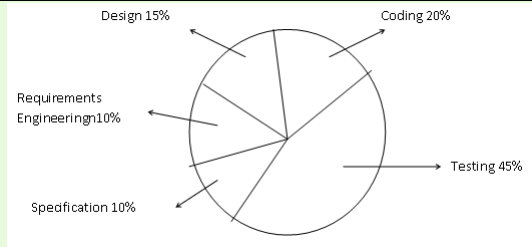


Figure 3.2:

From this data a very clear trend emerges, the so called 40-20-40 rule; only 20% of the effort is spent on actually programming (coding) the system, while the preceding phases (requirements engineering and design) and testing each consume about 40% of the total effort.



Revision Questions

Example . The software life cycle can be said to consist of a series of phases. The classical model is referred to as the waterfall model. Which phase may be defined as “The concept is explored and refined, and the client’s requirements are elicited?”

- (a) Requirements
- (b) Specification
- (c) Design
- (d) Implementation
- (e) Integration.

Solution: Requirements ☐

EXERCISE 1.  In the maintenance phase the product must be tested against previous test cases. This is known as _____



testing.

- (a) Unit
- (b) Integration
- (c) Regression
- (d) Module
- (e) Beta.

References and Additional Reading Materials

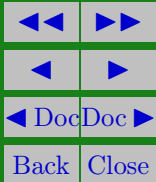
1. Hans van Vliet (2005). Software Engineering: Principles and Practice (2nd ed.). John Wiley & Sons, Inc. ISBN: 0-471-97508-7
2. Ian Sommerville, Pete Sawyer(2005). Requirements Engineering: A Good Practice Guide. ISBN: 0-471-97444-7. John Wiley & Sons, Inc



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3. Gerald Kotonya, Ian Sommerville (1998). Requirements Engineering: Processes and Techniques. John Wiley & Sons, Inc. ISBN: 0-471-97208-8

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Solutions to Exercises

Exercise 1.

Regression

Exercise 1

